

THIRUVANANTHAPURAM, India

WMO: 433710

Lat: 8.508N

Lon: 76.958E

Elev: 64

StdP: 100.56

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.3	22.8	16.5	11.8	28.7	17.8	12.9	28.0	6.8	28.7	6.0	29.0	0.9	50	0.312

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	7.2	34.2	26.2	33.6	26.0	33.1	25.8	28.0	31.9	27.6	31.5	27.2	31.1	2.0	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.0	22.9	30.4	26.6	22.3	30.0	26.2	21.8	29.6	90.2	32.1	88.3	31.7	86.7	31.3	30.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.1	4.2	3.4	DB	20.3	35.9	1.8	1.4	19.0	36.9	18.0	37.8	17.0	38.6	15.7	39.6	(4)
(5)			WB	17.9	29.1	1.1	0.8	17.1	29.6	16.5	30.1	15.9	30.5	15.1	31.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	27.9	27.6	28.3	29.2	29.3	28.9	27.4	27.0	27.1	27.4	27.3	27.5	(6)	
(7)		DBStd	1.30	0.84	0.84	0.93	1.17	1.32	1.23	1.00	0.98	0.97	1.05	0.93	0.87	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD10.0	6519	546	513	596	580	586	522	526	531	522	538	519	541	(10)
(11)		CDD18.3	3477	287	279	337	330	327	272	267	273	272	280	269	283	(11)
(12)		CDH23.3	36058	2892	3058	4060	4046	3889	2669	2408	2530	2608	2616	2510	2773	(12)
(13)		CDH26.7	13088	1131	1269	1809	1833	1619	812	631	696	775	783	761	970	(13)
(14)	Wind	WSAvg	1.2	0.7	0.9	1.0	1.1	1.4	1.6	2.0	1.9	1.5	0.9	0.6	0.6	(14)
(15)	Precipitation	PrecAvg	1780	19	24	38	119	203	302	203	150	180	276	207	64	(15)
(16)		PrecMax	2420	162	128	313	309	593	872	421	526	499	656	722	344	(16)
(17)		PrecMin	1127	0	0	0	5	7	58	47	21	4	17	15	0	(17)
(18)		PrecStd	319	30	34	48	69	127	167	90	100	136	146	130	69	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.3	34.2	35.0	35.1	34.3	33.1	31.7	32.0	32.7	32.8	32.6	33.0	(19)
(20)			MCWB	23.9	24.2	25.8	27.0	27.4	26.6	26.3	25.7	25.8	25.6	25.9	24.7	(20)
(21)		2%	DB	32.6	33.4	34.1	34.2	33.4	31.8	31.0	31.2	31.8	31.9	31.8	32.2	(21)
(22)			MCWB	24.0	24.2	25.8	27.0	27.1	26.6	26.0	25.7	25.8	25.7	25.7	24.7	(22)
(23)		5%	DB	32.1	32.7	33.5	33.5	32.8	31.1	30.4	30.6	31.1	31.2	31.2	31.7	(23)
(24)			MCWB	24.0	24.2	25.7	26.9	26.9	26.4	25.9	25.7	25.7	25.8	25.6	24.7	(24)
(25)		10%	DB	31.5	32.1	32.8	32.8	32.1	30.4	29.7	30.0	30.4	30.4	30.4	31.0	(25)
(26)			MCWB	24.0	24.3	25.6	26.7	26.8	26.2	25.7	25.6	25.6	25.7	25.5	24.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.6	26.7	28.1	28.7	28.7	28.0	27.2	27.1	27.6	27.4	27.2	27.1	(27)
(28)			MCD B	31.1	31.7	32.6	32.8	32.7	30.9	30.3	30.1	30.5	30.5	30.8	30.9	(28)
(29)		2%	WB	25.8	26.2	27.2	28.0	27.9	27.2	26.6	26.5	26.6	26.7	26.6	26.4	(29)
(30)			MCD B	30.2	31.0	32.0	32.2	31.8	30.3	29.7	29.7	30.0	30.0	30.2	30.2	(30)
(31)		5%	WB	25.4	25.7	26.7	27.5	27.5	26.8	26.2	26.1	26.2	26.3	26.2	25.9	(31)
(32)			MCD B	29.7	30.3	31.4	31.8	31.3	29.9	29.3	29.4	29.7	29.6	29.7	29.7	(32)
(33)		10%	WB	25.0	25.4	26.4	27.2	27.2	26.5	25.9	25.7	25.8	25.9	25.8	25.5	(33)
(34)			MCD B	29.2	29.7	31.0	31.4	30.8	29.4	29.0	28.9	29.2	29.2	29.1	29.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.4	8.4	7.9	7.2	6.3	5.6	5.5	5.7	6.1	6.3	6.4	7.5	(35)
(36)			MCDBR	8.9	8.8	8.4	7.8	7.0	6.5	6.3	6.5	7.0	7.2	7.5	8.4	(36)
(37)			MCWBR	3.4	3.1	2.8	2.8	2.6	2.7	2.7	2.4	2.6	2.7	3.0	3.3	(37)
(38)		5% WB	MCDBR	7.9	7.8	7.8	7.3	6.5	5.9	5.9	5.9	6.3	6.5	6.7	7.2	(38)
(39)			MCWBR	3.2	2.9	2.9	3.1	2.9	2.8	2.8	2.6	2.7	2.9	3.1	3.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.475	0.499	0.568	0.577	0.577	0.530	0.549	0.525	0.483	0.518	0.528	0.496	(40)	
(41)		taud	2.196	2.123	1.953	1.953	1.960	2.103	2.030	2.112	2.249	2.136	2.086	2.156	(41)	
(42)		Ebn at Noon	839	830	777	760	743	769	758	790	834	804	784	809	(42)	
(43)		Edh at Noon	147	162	194	192	186	159	172	162	142	158	163	150	(43)	
(44)	All-Sky Solar Radiation	RadAvg	5.45	6.13	6.32	5.97	5.45	5.00	5.11	5.40	5.62	5.10	4.42	4.76	(44)	
(45)		RadStd	0.25	0.24	0.37	0.29	0.43	0.41	0.34	0.38	0.39	0.34	0.28	0.30	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.28	N/A	N/A	+0.55	+0.60	+0.55	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		+0.28	N/A	N/A	+0.55	+0.60	+0.55	N/A	N/A	N/A	N/A

Nomenclature: See separate page